

CSE400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple RVs
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1 Statistical Representation of Random Vectors

For multiple random variables $X_1, X_2, \dots, X_N$, we utilize a compact vector representation to manage their interdependencies.

1.1 Random Vector Definition

A random vector $\underline{X}$ is a column vector containing N random variables:

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

1.2 Mean Vector

The mean vector $\underline{\mu}_X$ represents the average value of the random vector, where each element corresponds to the expected value of an individual random variable:

$$\underline{\mu}_X = E[\underline{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

1.3 Correlation Matrix

The correlation matrix R_{XX} captures the correlation between multiple random variables.

Definition: $R_{XX} = E[\underline{X}\underline{X}^T]$.

Element Representation: For $i, j \leq N$, the element $R_{XX}(i, j) = E[X_i X_j]$.

Matrix Structure:

$$R_{XX} = \begin{bmatrix} E[X_1 X_1] & E[X_1 X_2] & \dots & E[X_1 X_N] \\ E[X_2 X_1] & E[X_2 X_2] & \dots & E[X_2 X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_N X_1] & E[X_N X_2] & \dots & E[X_N X_N] \end{bmatrix}_{N \times N}$$

Property: R_{XX} is a symmetric matrix.

1.4 Covariance Matrix

The covariance matrix C_{XX} serves as a "relationship map" for the random vector.

Definition: $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.

Practical Structure:

$$C_{XX} = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \dots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Property: Due to the fact that $\text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i)$, C_{XX} is symmetric.

2 Linear Transformation of Random Vectors

We consider a linear model where an input random vector $\underline{X}$ is transformed into an output vector $\underline{Y}$.

2.1 The Linear Model

The transformation is defined as:

$$\underline{Y} = \underline{A}\underline{X} + \underline{b}$$

- $\underline{A}$: Transformation matrix (scales and rotates data).
- $\underline{b}$: Constant vector (shifts/translates the center).
- $\underline{X}$: Input random vector.

For N -dimensional vectors, this expands to:

$$\begin{aligned} Y_1 &= a_{11}X_1 + a_{12}X_2 + \dots + a_{1N}X_N + b_1 \\ &\vdots \\ Y_N &= a_{N1}X_1 + a_{N2}X_2 + \dots + a_{NN}X_N + b_N \end{aligned}$$

3 Properties of Transformed Vectors

When $\underline{Y} = \underline{A}\underline{X} + \underline{b}$, the statistical properties of $\underline{Y}$ are derived as follows:

3.1 Mean of Transformed Vector ($\underline{\mu}_Y$)

Derivation:

1. Substitute $\underline{Y} = \underline{A}\underline{X} + \underline{b}$ into the expectation: $\underline{\mu}_Y = E[\underline{A}\underline{X} + \underline{b}]$.
2. Apply linearity of expectation: $E[\underline{A}\underline{X}] + E[\underline{b}]$.
3. **Result:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X + \underline{b}$.

3.2 Covariance of Transformed Vector (C_{YY})

Key Observation: Deviation from the mean is independent of shift $\underline{b}$:

$$\underline{Y} - \underline{\mu}_Y = (\underline{AX} + \underline{b}) - (\underline{A}\underline{\mu}_X + \underline{b}) = \underline{A}(\underline{X} - \underline{\mu}_X)$$

Derivation:

1. $C_{YY} = E[(\underline{Y} - \underline{\mu}_Y)(\underline{Y} - \underline{\mu}_Y)^T]$.
2. Substitute the observation: $C_{YY} = E[\underline{A}(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T \underline{A}^T]$.
3. **Simplified Result:** $C_{YY} = \underline{A}C_{XX}\underline{A}^T$.

3.3 Correlation of Transformed Vector (R_{YY})

Derivation:

1. $R_{YY} = E[\underline{Y}\underline{Y}^T] = E[(\underline{AX} + \underline{b})(\underline{AX} + \underline{b})^T]$.
2. Expansion of terms: $R_{YY} = E[\underline{AXX}^T \underline{A}^T + \underline{AX}\underline{b}^T + \underline{bX}^T \underline{A}^T + \underline{bb}^T]$.
3. Applying expectation: $R_{YY} = \underline{A}E[\underline{XX}^T]\underline{A}^T + \underline{A}E[\underline{X}]\underline{b}^T + \underline{b}E[\underline{X}^T]\underline{A}^T + \underline{bb}^T$.
4. **Final Result:** $R_{YY} = \underline{A}R_{XX}\underline{A}^T + \underline{A}\underline{\mu}_X\underline{b}^T + \underline{b}\underline{\mu}_X^T \underline{A}^T + \underline{bb}^T$.

4 Numerical Example

Given: $\underline{Y} = \underline{AX}$ where:

$$\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

Solution (a): Mean Vector of $\underline{Y}$

Using $\underline{\mu}_Y = \underline{A}\underline{\mu}_X$:

$$\underline{\mu}_Y = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} (1)(2) + (0)(-1) + (-1)(1) \\ (0)(2) + (-1)(-1) + (1)(1) \\ (-1)(2) + (1)(-1) + (0)(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

5 Case Study: Smart City Traffic & Weather (Ahmedabad)

We model a city's congestion using five random variables: Rainfall, Temperature, Traffic Volume, Speed, and Pollution.

Assumption: The entire system behaves as a Joint Gaussian Distribution, defined by its Mean (typical conditions) and Covariance (interdependencies).

Inferences from Covariance Matrix:

- $\uparrow$ Rainfall $\implies \uparrow$ Traffic (Congestion).

- $\uparrow$ Rainfall $\implies$ $\downarrow$ Speed.
- $\uparrow$ Traffic $\implies$ $\uparrow$ Pollution.
- $\uparrow$ Temperature $\implies$ Pollution disperses.

Transformation: A city defined "Congestion Index" is a linear transformation of these variables, allowing for predictive modeling of city conditions.